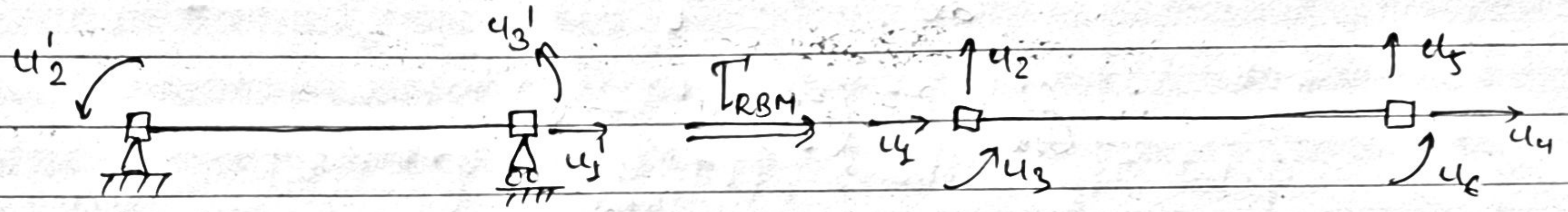
Homework : 2

① For the two-node linear-elastic Euler-Bernoulli beam element shown below (EI and EA constant), derive the 3x3 stiffness in the element basic local coordinate system shown - note that you need to solve the governing differential equations with the given boundary conditions. Then, use the rigid body motion transformation derived in the class to find the 6x6 element stiffness matrix. Compare the derived stiffness matrix with what you found in the first problem of HW #1.



→ Sol.

ⓐ For Axial Behaviour,

Equillibrium: $\frac{dP(x)}{dx} = 0$ — (I)

Compatibility: $\varepsilon_x = \frac{du}{dx}$ — (II)

Constitutive: $\sigma_x = E\varepsilon_x$

$\Rightarrow P = EA\,\varepsilon_x$ — (III)

$\Rightarrow EA\frac{d^2u}{dx^2} = 0$ — (IV)

$\therefore u(x) = c_1 x + c_2$ — (V)

Boundary Conditions:

At $x=0$, $u(0)=0 \Rightarrow C_2=0$

At $x=L$, $u(L)=u_j' \Rightarrow C_1 L = u_j' \Rightarrow C_1 = u_j'/L$

Also,

$$P = EA\frac{du}{dx} = EAC_1 = EA\frac{u_j'}{L}$$

$$\therefore f_{jx}' = P = \left(\frac{EA}{L}\right) u_j' \quad \text{— (VI)}$$

(B) For Bending Behaviour

Equilibrium : $\frac{dM}{dx} = V$, $\frac{dV}{dx} = w(x)$ — (VII)

Constitutive : $K = \frac{M}{EI}$ — (VIII)

Compatibility : $\phi = \frac{dy}{dx}$ — (IX)

$$K = \frac{d\phi}{dx} \quad \text{— (X)}$$

for $w(x)=0$ and EI Constant,

$$\frac{d^4y}{dx^4} = 0$$

$$\therefore y(x) = C_3x^3 + C_4x^2 + C_5x + C_6 \quad \text{— (XI)}$$

$$\Rightarrow \phi(x) = 3C_3x^2 + 2C_4x + C_5 \quad \text{(XII)}$$

Boundary Conditions:

At $x=0$, $y(0)=0 \Rightarrow C_6 = 0$

At $x=0$, $\phi(0) = u_2' \Rightarrow C_5 = u_2'$

At $x=L$, $y(L)=0 \Rightarrow C_3L^2 + C_4L + u_2' = 0$

At $x=L$, $\alpha(L) = u_3' \Rightarrow 3C_3L^2 + 2C_4L + u_2' = u_3'$

Solving gives,

$$C_3 = \frac{u_2'+u_3'}{2}, \quad C_4 = -\frac{(2u_2'+u_3')}{L}$$

$$\therefore y(x) = \left(\frac{u_2'+u_3'}{L^2}\right)x^3 - \left(\frac{2u_2'+u_3'}{L}\right)x^2 + u_2'x \quad \text{(XIII)}$$

$$M(x) = EI\frac{d^2y}{dx^2} = \left(\frac{u_2'+u_3'}{L^2}\right)EI(6x) - \left(\frac{2u_2'+u_3'}{L}\right)2EI \quad \text{(XIV)}$$

At $x=0$, $m_2' = -m = \frac{EI}{L}(4u_2' + 2u_3')$

At $x=L$, $m_3' = m = EI(6L)\left(\frac{u_2'+u_3'}{L^2}\right) - 2EI\left(\frac{2u_2'+u_3'}{L}\right)$

$$= \frac{EI}{L}(2u_2' + 4u_3')$$

$\therefore$ Basic stiffness Matrix, $[K]_{basic} = \begin{bmatrix} AE/L & 0 & 0 \\ 0 & 4EI/L & 2EI/L \\ 0 & 2EI/L & 4EI/L \end{bmatrix}$

$$T_{RBM} = \begin{bmatrix} -1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1/L & 1 & 0 & -1/L & 0 \\ 0 & 1/L & 0 & 0 & -1/L & 1 \end{bmatrix}$$

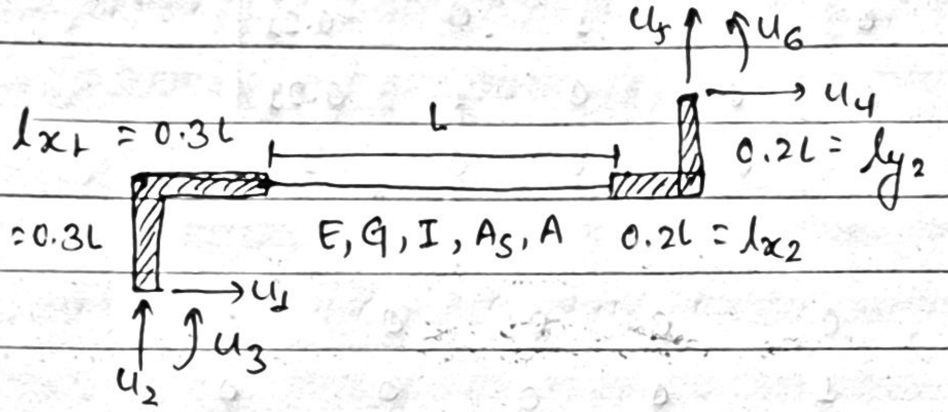
Global Stiffness Matrix, $[K]_{global} = T_{RBM}^{T} \; [K]_{basic} \; T_{RBM}$

$$= \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1/L & 1/L \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & -1/L & -1/L \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -AE/L & 0 & 0 & AE/L & 0 & 0 \\ 0 & 6EI/L^2 & 4EI/L & 0 & -6EI/L^2 & 2EI/L \\ 0 & 6EI/L^2 & 2EI/L & 0 & -6EI/L^2 & 4EI/L \end{bmatrix}$$

$$\Rightarrow [K]_{global} = \begin{bmatrix} AE/L & 0 & 0 & -AE/L & 0 & 0 \\ 0 & 12EI/L^3 & 6EI/L^2 & 0 & -12EI/L^3 & 6EI/L^2 \\ 0 & 6EI/L^2 & 4EI/L & 0 & -6EI/L^2 & 2EI/L \\ -AE/L & 0 & 0 & AE/L & 0 & 0 \\ 0 & -12EI/L^3 & -6EI/L^2 & 0 & 12EI/L^3 & -6EI/L^2 \\ 0 & 6EI/L^2 & 2EI/L & 0 & -6EI/L^2 & 4EI/L \end{bmatrix}$$

This is the same stiffness matrix derived in HW #1, problem #1.

② For the beam element with rigid end zones shown in the figure, derive the 6X6 element stiffness matrix. Assume linear-elastic Timoshenko beam theory and use the element stiffness matrix derived in the class.



→ Sol.

we have,

Local stiffness Matrix = $[K]_{local}$

$$[K]_{local} = \frac{EI}{l(1+\phi)} \begin{bmatrix} \frac{A(1+\phi)}{I} & 0 & 0 & -\frac{A(1+\phi)}{I} & 0 & 0 \\ 0 & 12/l^2 & 6/l & 0 & -12/l^2 & 6/l \\ 0 & 6/l & 4+\phi & 0 & -6/l & 2-\phi \\ -\frac{A(1+\phi)}{I} & 0 & 0 & \frac{A(1+\phi)}{I} & 0 & 0 \\ 0 & -12/l^2 & -6/l & 0 & 12/l^2 & -6/l \\ 0 & 6/l & 2-\phi & 0 & -6/l & 4+\phi \end{bmatrix}$$

where, $\phi = \frac{12EI}{GA_sl^2}$ = shear correction term

$$T_{REZ} = \begin{bmatrix} 1 & 0 & -0.3l & 0 & 0 & 0 \\ 0 & 1 & 0.3l & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & -0.2l \\ 0 & 0 & 0 & 0 & 1 & 0.2l \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

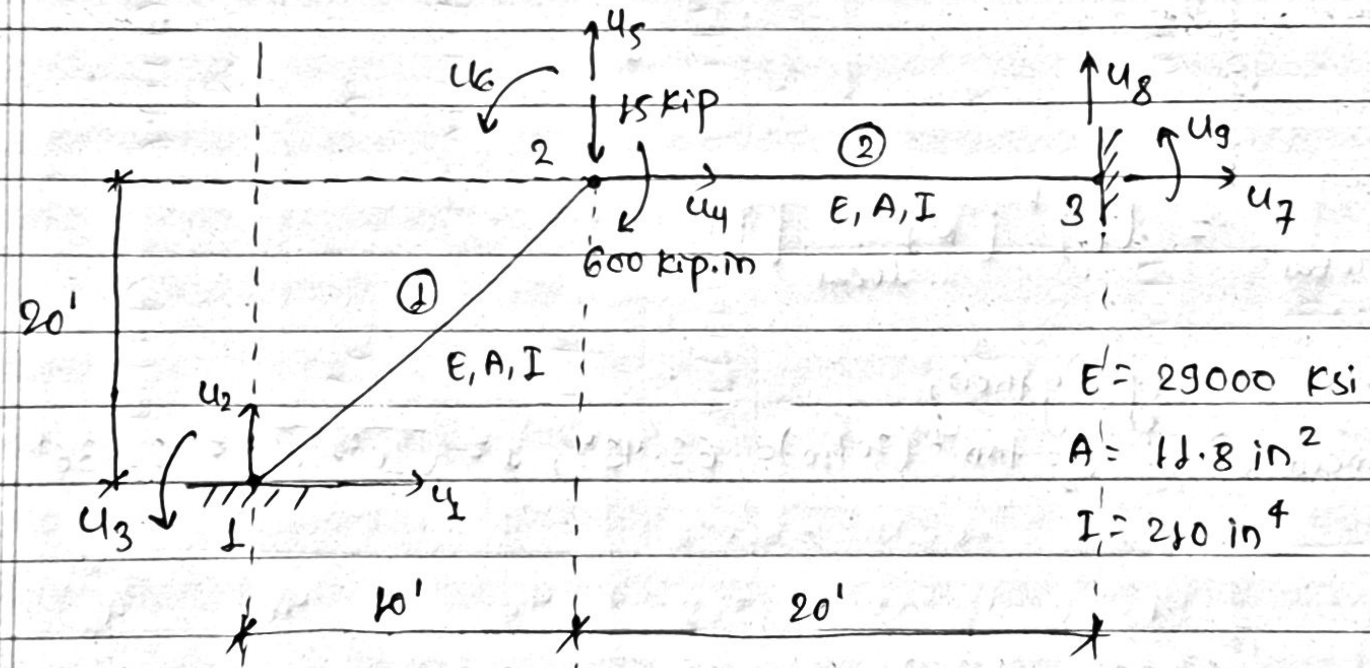
$$T_{REZ}^{T} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ -0.3l & 0.3l & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -0.2l & 0.2l & 1 \end{bmatrix}$$

$$[K]_{global} = T_{REZ}^{T}\,[K]_{local}\,T_{REZ}$$

$$[K]_{local}\,T_{REZ} = \frac{EI}{l(1+\phi)} \begin{bmatrix} \frac{A(1+\phi)}{I} & 0 & -\frac{0.3A(1+\phi)l}{I} & -\frac{A(1+\phi)}{I} & 0 & \frac{0.2A(1+\phi)l}{I} \\ 0 & 12/l^2 & 9.6/l & 0 & -12/l^2 & 3.6/l \\ 0 & 6/l & 5.8+\phi & 0 & -6/l & 0.8-\phi \\ -\frac{A(1+\phi)}{I} & 0 & \frac{0.3A(1+\phi)l}{I} & \frac{A(1+\phi)}{I} & 0 & -\frac{0.2A(1+\phi)l}{I} \\ 0 & -12/l^2 & -9.6/l & 0 & 12/l^2 & -3.6/l \\ 0 & 6/l & 3.8-\phi & 0 & -6/l & 2.8+\phi \end{bmatrix}$$

$$[K]_{global} = \frac{EI}{l(1+\phi)} \begin{bmatrix} \frac{A(1+\phi)}{I} & 0 & -\frac{0.3Al(1+\phi)}{I} & -\frac{A(1+\phi)}{I} & 0 & \frac{0.2Al(1+\phi)}{I} \\ 0 & 12/l^2 & 9.6/l & 0 & -12/l^2 & 3.6/l \\ -\frac{0.3Al(1+\phi)}{I} & 9.6/l & \frac{0.09Al^2(1+\phi)}{I} + 8.68 + \phi & \frac{0.3Al(1+\phi)}{I} & -9.6/l & -\frac{0.06Al^2(1+\phi)}{I} + 1.88 - \phi \\ -\frac{A(1+\phi)}{I} & 0 & \frac{0.3Al(1+\phi)}{I} & \frac{A(1+\phi)}{I} & 0 & -\frac{0.2Al(1+\phi)}{I} \\ 0 & -12/l^2 & -9.6/l & 0 & 12/l^2 & -3.6/l \\ \frac{0.2Al(1+\phi)}{I} & 3.6/l & -\frac{0.06Al^2(1+\phi)}{I} + 1.88 - \phi & -\frac{0.2Al(1+\phi)}{I} & -3.6/l & \frac{0.04Al^2(1+\phi)}{I} + 2.08 + \phi \end{bmatrix}$$

(3) Assuming Euler-Bernoulli beam theory, derive the stiffness matrix for the following structure, and solve for the nodal displacements. Then determine the state of each element, i.e., nodal displacements and forces in the corresponding element local coordinate system. Finally, to make sure you have solved the problem correctly, check the equillibrium at node 2. Feel free to use MATLAB or other software for matrix calculations.



→ Sol^n.

$$\text{Transformation Matrix, } [T] = \begin{bmatrix} c & s & 0 & 0 & 0 & 0 \\ -s & c & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & c & s & 0 \\ 0 & 0 & 0 & -s & c & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$[K]_{local} = \begin{bmatrix} AE/l & 0 & 0 & -AE/l & 0 & 0 \\ 0 & 12EI/l^3 & 6EI/l^2 & 0 & -12EI/l^3 & 6EI/l^2 \\ 0 & 6EI/l^2 & 4EI/l & 0 & -6EI/l^2 & 2EI/l \\ -AE/l & 0 & 0 & AE/l & 0 & 0 \\ 0 & -12EI/l^3 & -6EI/l^2 & 0 & 12EI/l^3 & -6EI/l^2 \\ 0 & 6EI/l & 2EI/l & 0 & -6EI/l^2 & 4EI/l \end{bmatrix}$$

$$[K]_{global} = [T]^T [K]_{local} [T]$$

Using python,
for element ①, $\theta_1 = \tan^{-1}(20/10) = 63.43°$, $l = \sqrt{10^2+20^2} \times 12 = 268.33''$

$$[K]_{global\,①} = 10^2 \begin{bmatrix} 2.58 & 5.09 & -4.54 & -2.58 & -5.09 & -4.54 \\ & 10.21 & 2.27 & -5.09 & -10.21 & 2.27 \\ & & 907.87 & 4.54 & -2.27 & 453.94 \\ & & & 2.58 & 5.09 & 4.54 \\ & & & & 10.21 & -2.27 \\ & \text{Sym.} & & & & 907.87 \end{bmatrix}$$

for element ②, $\theta_2 = 0°$, $l = 20 \times 12 = 240''$

$$[K]_{global\,②} = 10^2 \begin{bmatrix} 14.26 & 0 & 0 & -14.26 & 0 & 0 \\ & 0.053 & 6.34 & 0 & -0.053 & 6.34 \\ & & 1015.0 & 0 & -6.34 & 507.5 \\ & & & 14.26 & 0 & 0 \\ & & & & 0.053 & -6.34 \\ & \text{Sym.} & & & & 1015.0 \end{bmatrix}$$

Global stiffness Matrix,

$$[K] = 10^2 \begin{array}{c} \\ \left[\begin{array}{ccccccccc} 2.58 & 5.09 & -4.54 & -2.58 & -5.09 & -4.54 & 0 & 0 & 0 \\ & 10.21 & 2.27 & -5.09 & -10.21 & 2.27 & 0 & 0 & 0 \\ & & 907.87 & 4.54 & -2.27 & 453.94 & 0 & 0 & 0 \\ & & & 16.84 & 5.09 & 4.54 & -14.26 & 0 & 0 \\ & & & & 10.26 & 4.07 & 0 & -0.053 & 6.34 \\ & & & & & 1922.87 & 0 & -6.34 & 507.5 \\ & & & & & & 14.26 & 0 & 0 \\ & & & & & & & 0.053 & -6.34 \\ & \text{Sym.} & & & & & & & 1015.0 \end{array}\right] \end{array}$$

	u_1	u_2	u_3	u_4	u_5	u_6	u_7	u_8	u_9	
	2.58	5.09	-4.54	-2.58	-5.09	-4.54	0	0	0	u_1
		10.21	2.27	-5.09	-10.21	2.27	0	0	0	u_2
			907.87	4.54	-2.27	453.94	0	0	0	u_3
				16.84	5.09	4.54	-14.26	0	0	u_4
					10.26	4.07	0	-0.053	6.34	u_5
						1922.87	0	-6.34	507.5	u_6
							14.26	0	0	u_7
								0.053	-6.34	u_8
	Sym.								1015.0	u_9

$$\{f\} = \begin{Bmatrix} f_1 \\ f_2 \\ m_3 \\ 0 \\ -15 \\ -600 \\ f_7 \\ f_8 \\ m_9 \end{Bmatrix} \qquad \{u\} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ u_4 \\ u_5 \\ u_6 \\ 0 \\ 0 \\ 0 \end{Bmatrix}$$

To solve for displacements,

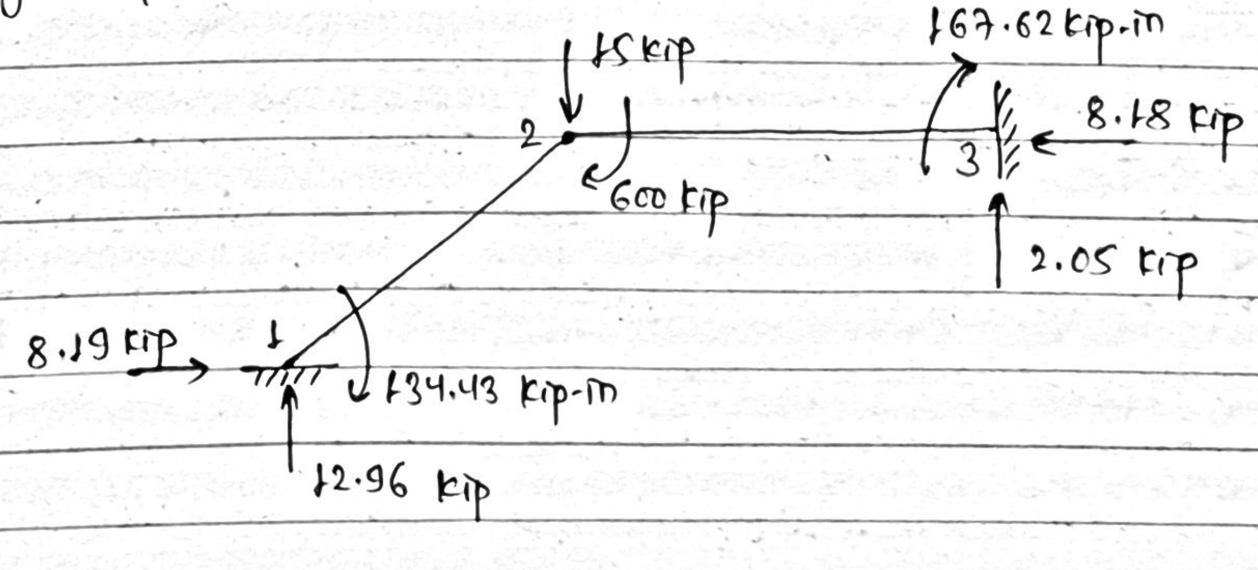
$$\begin{Bmatrix} 0 \\ -15 \\ 600 \end{Bmatrix} = 10^2 \begin{bmatrix} 16.84 & 5.09 & 4.54 \\ 5.09 & 10.26 & 4.07 \\ 4.54 & 4.07 & 1922.87 \end{bmatrix} \begin{Bmatrix} u_4 \\ u_5 \\ u_6 \end{Bmatrix}$$

$$\begin{Bmatrix} u_4 \\ u_5 \\ u_6 \end{Bmatrix} = 10^{-2} \begin{Bmatrix} 0.574 \\ -1.624 \\ -0.310 \end{Bmatrix}$$

$$\begin{Bmatrix} f_1 \\ f_2 \\ m_3 \\ f_7 \\ f_8 \\ m_9 \end{Bmatrix} = 10^{0} \begin{bmatrix} -2.58 & -5.09 & -4.54 \\ -5.09 & -10.28 & 2.27 \\ 4.54 & -2.27 & 453.94 \\ -14.26 & 0 & 0 \\ 0 & -0.053 & -6.34 \\ 0 & 6.34 & 507.5 \end{bmatrix} \begin{Bmatrix} 0.574 \\ -1.624 \\ -0.310 \end{Bmatrix}$$

$$= \begin{Bmatrix} 8.19 \\ 12.96 \\ -134.43 \\ -8.18 \\ 2.05 \\ -167.62 \end{Bmatrix}$$; force in kip, moment in kip-m

for equillibrium at Node 2,



$\Sigma M_2 = 134.43 + 600 + 167.62 + 12.96 \times 10 \times 12 - 2.05 \times 20 \times 12 - 8.19 \times 20 \times 12$

$= -0.35 \approx 0$ OKAY